

Lecture 9: Likelihood for Censored Survival Data

BSDS Third Year Students

First Semester, AY 2026–27

Outline of Today's Lecture

From censoring to likelihood

Likelihood contributions

Exponential model

Weibull model

Gamma and other parametric models

Maximum likelihood with censoring

Key takeaway messages

- ▶ Event time: T .
- ▶ Survival function: $S(t) = \mathbb{P}(T > t)$.
- ▶ Density: $f(t)$, when it exists.
- ▶ Hazard function: $h(t) = f(t)/S(t)$.
- ▶ Right-censored observation: we know $T > c$, not $T = c$.

Today we ask: **how should exact and censored observations enter the likelihood?**

Let T be the true event time and let θ denote the parameter of a survival model.

Observed information	Meaning	Likelihood contribution
Exact event at t	$T = t$	$f(t; \theta)$
Right censored at c	$T > c$	$S(c; \theta)$
Left censored at c	$T \leq c$	$F(c; \theta)$
Interval censored in $(a, b]$	$a < T \leq b$	$S(a; \theta) - S(b; \theta)$

A small dataset from Lecture IX



Suppose we observe the following data:

Subject	Observed time u_i	Status δ_i	Interpretation
1	4	1	event at 4
2	6	0	event-free beyond 6
3	7	1	event at 7
4	9	0	event-free beyond 9
5	10	1	event at 10

Here

$\delta_i = 1$ means exact failure, $\delta_i = 0$ means right censoring.

What should subject 2 contribute to the likelihood: $f(6; \theta)$ or $S(6; \theta)$?

What subject 2 contributes



For subject 2, the observation is

$$(u_2, \delta_2) = (6, 0).$$

This means

$$T_2 > 6.$$

Therefore the probability of observing this information is

$$\mathbb{P}_\theta(T_2 > 6) = S(6; \theta).$$

$$U_i = \min(T_i, C_i),$$
$$\delta_j = \mathbb{I}(T_j \leq C_j).$$
$$\delta_j = 1 \implies U_j = T_j \quad \text{is an exact event time,}$$

$$\delta_i = 0 \implies U_i = C_i \quad \text{and} \quad T_i > C_i.$$

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Suppose subject i has an exact event time:

$$\delta_i = 1, \quad U_i = T_i = u_i.$$

The likelihood contribution is the density at the event time:

$$L_i(\theta) = f(u_i; \theta).$$

Using $f(t) = h(t)S(t)$, this can also be written as

$$L_i(\theta) = h(u_i; \theta)S(u_i; \theta).$$

Suppose subject i is right censored:

$$\delta_i = 0, \quad U_i = C_i = u_i.$$

The information is

$$T_i > u_i.$$

Therefore the likelihood contribution is

$$L_i(\theta) = \mathbb{P}_\theta(T_i > u_i) = S(u_i; \theta).$$

Right-censoring contributes a survival probability, not a density.

Combining exact and right-censored observations



For one observation (u_i, δ_i) , the two cases can be combined as

$$L_i(\theta) = \{f(u_i; \theta)\}^{\delta_i} \{S(u_i; \theta)\}^{1-\delta_i}.$$

Equivalently, since $f = hS$,

$$L_i(\theta) = \{h(u_i; \theta)\}^{\delta_i} S(u_i; \theta).$$

For independent observations,

$$L(\theta) = \prod_{i=1}^n \{f(u_i; \theta)\}^{\delta_i} \{S(u_i; \theta)\}^{1-\delta_i}.$$

This is the standard likelihood for independent right-censored survival data ([Cox and Oakes, 1984](#); [Collett, 2015](#)).

Log-likelihood form



Taking logarithms gives

$$\ell(\theta) = \sum_{i=1}^n [\delta_i \log f(u_i; \theta) + (1 - \delta_i) \log S(u_i; \theta)] .$$

Equivalently,

$$\ell(\theta) = \sum_{i=1}^n [\delta_i \log h(u_i; \theta) + \log S(u_i; \theta)] .$$

Why is the second form often conceptually useful?

Because it separates two parts:

- ▶ everyone must survive until their observed time;
- ▶ only uncensored subjects experience the event at the observed time.

What about the censoring mechanism?



Strictly speaking, the observed data depend on both T_i and C_i .

A standard assumption is **independent censoring**:

$$T_i \perp C_i$$

with the survival-model parameter θ appearing only in the distribution of T_i .

Then the part of the full likelihood involving θ is proportional to

$$\prod_{i=1}^n \{f(u_i; \theta)\}^{\delta_i} \{S(u_i; \theta)\}^{1-\delta_i}.$$

If censoring depends on the unobserved event time in a way not accounted for, this likelihood can be misleading.

Why independent censoring matters



Consider two situations.

Situation A: administrative censoring

A study ends on 31 December. Anyone event-free at that date is censored. This may be plausibly independent of the biological event time.

Situation B: informative dropout

Patients with worsening symptoms leave the study early. Then censoring may be associated with imminent failure.

The exponential survival model



The exponential model assumes a constant hazard rate:

$$T \sim \text{Exponential}(\lambda), \quad \lambda > 0.$$

Its main functions are

$$f(t; \lambda) = \lambda e^{-\lambda t}, \quad t > 0,$$

$$S(t; \lambda) = e^{-\lambda t},$$

$$h(t; \lambda) = \lambda.$$

The exponential model is simple and analytically useful, but its constant hazard assumption is often restrictive.

Exponential likelihood with censoring



For right-censored data,

$$L(\lambda) = \prod_{i=1}^n \{\lambda e^{-\lambda u_i}\}^{\delta_i} \{e^{-\lambda u_i}\}^{1-\delta_i}.$$

Simplifying,

$$L(\lambda) = \lambda^{\sum_i \delta_i} \exp \left(-\lambda \sum_{i=1}^n u_i \right).$$

Let

$$d = \sum_{i=1}^n \delta_i$$

be the number of observed failures. Then

$$L(\lambda) = \lambda^d \exp \left(-\lambda \sum_i u_i \right).$$

Exponential log-likelihood and score



The log-likelihood is

$$\ell(\lambda) = d \log \lambda - \lambda \sum_{i=1}^n u_i.$$

The score equation is

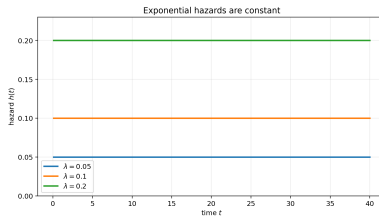
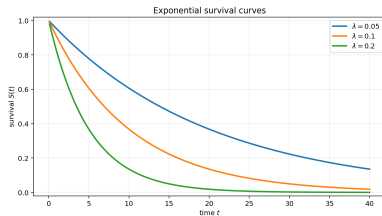
$$\frac{d\ell}{d\lambda} = \frac{d}{\lambda} - \sum_{i=1}^n u_i = 0.$$

Therefore,

$$\hat{\lambda} = \frac{d}{\sum_{i=1}^n u_i}.$$

The denominator uses the **total observed time on study**, including censored follow-up times.

Exponential survival and hazard: plots



Increasing λ makes survival drop faster, but the hazard is still flat over time. This is the key restriction of the exponential model.

An worked example: administrative censoring



Suppose a reliability experiment is stopped at 18 months. We observe

$$(u_i) = (3, 5, 6, 8, 11, 12, 15, 18),$$
$$(\delta_i) = (1, 0, 1, 1, 0, 1, 0, 1).$$

Here $\delta_i = 0$ means the unit was still working when last observed. Then

$$d = 5, \quad \sum_i u_i = 78.$$

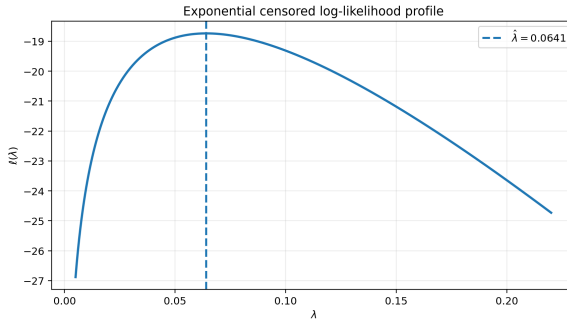
For the exponential model,

$$\hat{\lambda} = \frac{d}{\sum_i u_i} = \frac{5}{78} = 0.0641.$$

Log-likelihood profile for the example

For the same dataset,

$$\ell(\lambda) = 5 \log \lambda - 78\lambda.$$



The peak occurs at $\hat{\lambda} = 5/78$.

Why go beyond the exponential model?



The exponential model assumes

$$h(t) = \lambda,$$

so the instantaneous event rate is constant over time.

But many real phenomena have time-varying hazards.

- ▶ A machine may be more likely to fail as it ages.
- ▶ Early post-surgery risk may be high and then decline.
- ▶ A treatment effect may change over follow-up time.

The Weibull model is a simple parametric model that allows increasing or decreasing hazards.

The Weibull survival model



Use the shape-scale parameterization:

$$T \sim \text{Weibull}(k, \eta), \quad k > 0, \quad \eta > 0.$$

The survival function is

$$S(t; k, \eta) = \exp \left[- \left(\frac{t}{\eta} \right)^k \right].$$

The density is

$$f(t; k, \eta) = \frac{k}{\eta} \left(\frac{t}{\eta} \right)^{k-1} \exp \left[- \left(\frac{t}{\eta} \right)^k \right].$$

The hazard is

$$h(t; k, \eta) = \frac{k}{\eta} \left(\frac{t}{\eta} \right)^{k-1}.$$

Interpreting the Weibull shape parameter



For the Weibull model,

$$h(t; k, \eta) = \frac{k}{\eta} \left(\frac{t}{\eta} \right)^{k-1}.$$

Shape k	Hazard behavior	Interpretation
$k = 1$	constant	exponential special case
$k > 1$	increasing	aging, cumulative damage
$0 < k < 1$	decreasing	early high-risk period

Weibull MLE: one analytic simplification



For fixed k , differentiate $\ell(k, \eta)$ with respect to η .

The relevant terms are

$$-dk \log \eta - \sum_i \frac{u_i^k}{\eta^k}.$$

Setting the derivative equal to zero gives

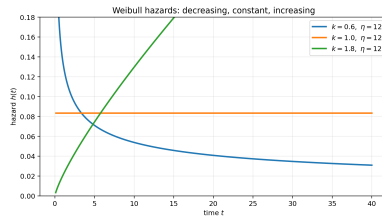
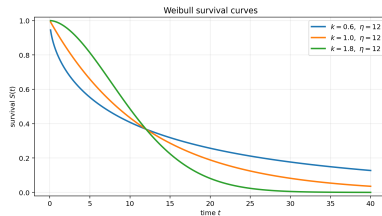
$$-\frac{dk}{\eta} + \frac{k}{\eta^{k+1}} \sum_i u_i^k = 0.$$

Hence

$$\hat{\eta}(k)^k = \frac{\sum_i u_i^k}{d}.$$

For Weibull MLE, η has a closed-form update if k is fixed, but k usually requires numerical solution.

Weibull survival and hazard: plots



The same scale $\eta = 12$ can produce decreasing, constant, or increasing hazards depending on k .

Worked example: evaluating a Weibull likelihood



Use the data from the administrative-censoring example:

$$(u_i) = (3, 5, 6, 8, 11, 12, 15, 18), \quad (\delta_i) = (1, 0, 1, 1, 0, 1, 0, 1).$$

Evaluate the Weibull log-likelihood at $k = 1.5, \eta = 12$:

$$\ell(k, \eta) = \sum_i \delta_i \log h(u_i; k, \eta) + \sum_i \log S(u_i; k, \eta).$$

Using

$$\log h(u_i) = \log(1.5) - \log(12) + 0.5 \log(u_i/12),$$

$$\log S(u_i) = -(u_i/12)^{1.5},$$

we get

$$\ell(1.5, 12) = -17.8411.$$

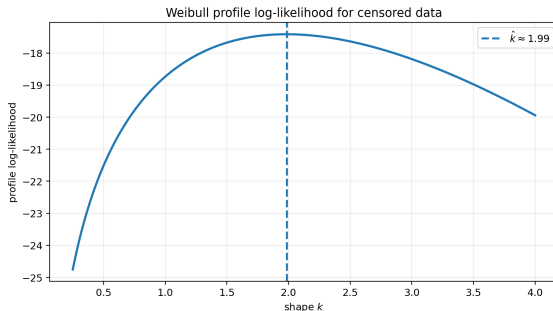
Worked example: profiling the Weibull shape



For fixed k , recall that

$$\hat{\eta}(k)^k = \frac{\sum_i u_i^k}{d}.$$

Thus we can reduce the two-dimensional optimization to a one-dimensional profile over k .



For this dataset, the profile maximum is approximately

$$\hat{k} \approx 1.99, \quad \hat{\eta} \approx 13.77.$$

The gamma model



A gamma survival model may be written as

$$T \sim \text{Gamma}(\alpha, \beta),$$

where $\alpha > 0$ is a shape parameter and $\beta > 0$ is a rate parameter.

The density is

$$f(t; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t}, \quad t > 0.$$

The survival function is

$$S(t; \alpha, \beta) = \mathbb{P}_{\alpha, \beta}(T > t).$$

Unlike the exponential model, the gamma survival function generally involves the incomplete gamma function.

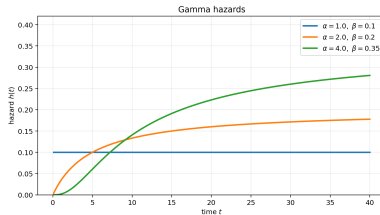
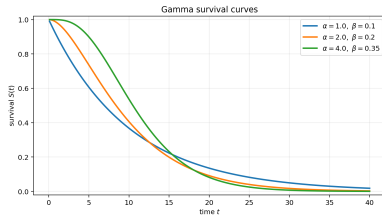
The censored likelihood is still conceptually simple:

$$L(\alpha, \beta) = \prod_{i=1}^n \{f(u_i; \alpha, \beta)\}^{\delta_i} \{S(u_i; \alpha, \beta)\}^{1-\delta_i}.$$

The log-likelihood is

$$\ell(\alpha, \beta) = \sum_i \delta_i \log f(u_i; \alpha, \beta) + \sum_i (1 - \delta_i) \log S(u_i; \alpha, \beta).$$

Gamma survival and hazard: plots



Gamma models are more flexible than exponential models, but their survival function and MLE usually need numerical routines.

Worked example: gamma likelihood contributions



Let $T \sim \text{Gamma}(\alpha, \beta)$ with $\alpha = 2$ and $\beta = 0.2$. Consider two observations:

$$(u_1, \delta_1) = (6, 1), \quad (u_2, \delta_2) = (11, 0).$$

The first is an exact failure, so its contribution is

$$f(6; 2, 0.2) = 0.0723, \quad \log f(6; 2, 0.2) = -2.6271.$$

The second is right-censored, so its contribution is

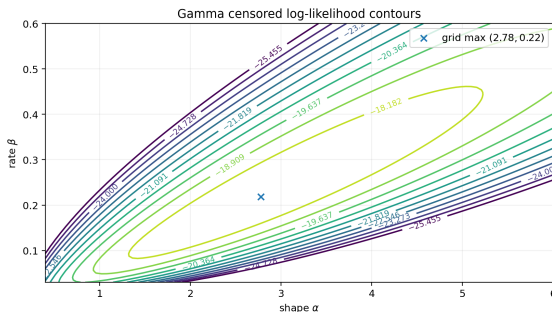
$$S(11; 2, 0.2) = 0.3546, \quad \log S(11; 2, 0.2) = -1.0368.$$

The same observed time enters as a density if $\delta = 1$, and as a survival probability if $\delta = 0$.

Worked example: gamma likelihood surface

For the administrative-censoring dataset, the censored gamma log-likelihood is

$$\ell(\alpha, \beta) = \sum_i \delta_i \log f(u_i; \alpha, \beta) + \sum_i (1 - \delta_i) \log S(u_i; \alpha, \beta).$$



A simple grid search gives an approximate maximum near

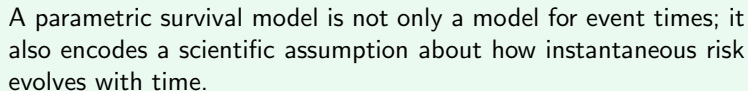
$$\hat{\alpha} \approx 2.78, \quad \hat{\beta} \approx 0.218.$$

Common parametric survival models



Model	Typical use	Hazard behavior	MLE with censoring
Exponential	first simple model	constant	often analytic
Weibull	reliability, medical studies	monotone	partly analytic, partly numerical
Gamma	flexible waiting time	non-monotone	numerical
Log-normal	skewed event times	non-monotone	numerical
Log-logistic	long-tailed survival	non-monotone	numerical

Parametric survival analysis is likelihood-based: choose a distribution for T , write f and S , then maximize the censored likelihood.



The general MLE problem



For a parametric family indexed by θ , the MLE is

$$\hat{\theta} = \arg \max_{\theta} \ell(\theta),$$

where

$$\ell(\theta) = \sum_i [\delta_i \log f(u_i; \theta) + (1 - \delta_i) \log S(u_i; \theta)].$$

Equivalently,

$$\ell(\theta) = \sum_i [\delta_i \log h(u_i; \theta) + \log S(u_i; \theta)].$$

The score function is

$$U(\theta) = \frac{\partial \ell(\theta)}{\partial \theta}.$$

The MLE usually solves

$$U(\hat{\theta}) = 0.$$

For censored data,

$$U(\theta) = \sum_i \left[\delta_i \frac{\partial}{\partial \theta} \log f(u_i; \theta) + (1 - \delta_i) \frac{\partial}{\partial \theta} \log S(u_i; \theta) \right].$$

The censored observations are not absent from the score. They contribute through $\partial \log S(u_i; \theta) / \partial \theta$.

Why censoring is not MCAR



In missing data language, MCAR means that the missingness indicator is unrelated to the observed and unobserved data values.

Right censoring is different:

$$\delta_i = 0 \Rightarrow T_i > U_i.$$

So the observed censoring status directly tells us something about the unobserved event time.

A right-censored event time is not merely an unknown blank cell. It is constrained to lie beyond the censoring time.

This is why censored observations should not be deleted under the logic of complete-case analysis.

Key takeaway messages



1. Exact failure at t contributes $f(t; \theta)$.
2. Right censoring at c contributes $S(c; \theta)$.
3. For right-censored observations,

$$L(\theta) = \prod_i f(u_i; \theta)^{\delta_i} S(u_i; \theta)^{1-\delta_i}.$$

4. Equivalently,

$$L(\theta) = \prod_i h(u_i; \theta)^{\delta_i} S(u_i; \theta).$$

5. Exponential MLE under censoring is $\hat{\lambda} = d / \sum_i u_i$.
6. Weibull and gamma models follow the same likelihood principle but usually require numerical optimization.

In the next survival lecture, we will move from parametric likelihoods to nonparametric estimation of the survival curve.

- ▶ risk sets;
- ▶ event times;
- ▶ product-limit idea;
- ▶ Kaplan–Meier estimator;
- ▶ survival curves and median survival.

Today: likelihood for censored data.

Next: estimating $S(t)$ without specifying a parametric distribution.

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